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NPS/002/001 – Technical Specification for Earthing Materials

1. Purpose

The purpose of this document is to detail the requirements of CE Electric UK for earthing materials.

This document supersedes the following documents, all copies of which should be destroyed.

Ref.	Version	Title
NPS/002/001	1	Specification for Earthing Materials

2. Scope

This section of the tender document describes the requirements of CE Electric UK with respect to earthing materials, which shall comply with the requirements of ENATS – Energy Networks Technical Association Specification 43-94: Earth Rods and their Connectors for installation in accordance with CE Electric of Code of Practise (CoP) IMP/010/011.

- Earth Rods
- Couplings
- Steel Drive Stud's
- Bulldog Grip Earth Rod Connectors
- Flat Soft Drawn Copper Strip
- Flat Aluminium Strip
- Plain Copper Braid

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3. Technical Requirements

3.1 Copper Bonded Steel Earth Rod

Copper bonded steel earth rods shall be used to provide effective permanent connection of the distribution network to the general mass of earth. They shall have a thread pitch of Unified National Coarse 2A (UNC – 2A). The steel rod shall meet the requirements of BS EN 10025 minimum grade S275 JR and coated with a copper bond including the threaded area. A minimum coating of 250 microns should be applied.

3.2 Earth Rod Couplings:

Couplings shall be used for the series connection of earth rods as detailed in 3.1. The coupling shall have a threaded bore to allow them to be applied over the threaded end of the rod. They shall have the same mechanical characteristics as the rod and provide an adequate electrical conductance that is the equivalent of the rod its self.

3.3 Steel Driving Stud.

Steel driving studs shall be used to provide a durable impact point for a hammer when installing earth rods. The driving head shall be capable of being used multiple times without deformation that would affect its performance.

3.4 Earth Rod Connector.

The connector shall be manufactured to cater for conductors in the range 25mm² to 120mm² in size. The connector shall be compatible with the earth rod and the conductor so as to give a mechanically robust and a good electrical connection.

3.5 High Conductivity Plain Copper Strip

Tinned copper strip shall be used to provide earthing interconnection between items of substation plant and also to connect substation plant to earth arrangements. It shall be manufactured to BS EN 13601 and be embossed every 300mm with the wording "Property of CE Electric UK".

3.6 Flat Aluminium Strip.

Aluminium strip shall be used to provide earthing interconnection between items of substation plant and also to connect substation plant to earth arrangements. It shall be manufactured to BS 2829 - 1350 and be embossed every 300mm with the wording "Property of CE Electric UK".

3.7 Plain Copper Braid.

Copper braid shall be used to provide earthing interconnection between items of substation plant and also to connect substation plant to earth arrangements. It shall be manufactured in accordance with BS EN 13602.

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4 References

4.1 External Documentation

The Proposed items shall comply with the relevant International standards, British Standard Specifications and all relevant Energy Networks Association Technical Specifications (ENATS) current at the time of tendering, except where varied by this standard. In respect the following documents are particularly relevant.

Reference	Title
PD 970	Specification for wrought steels for mechanical and allied engineering purposes.
BS 951	Electrical earthing. Clamps for earthing and bonding. Specification
BS 2898 (1970)	Specification for wrought aluminium and aluminium alloys for electrical purposes. Bars, extruded round tube and sections
ESI 43-94	Earth Rods and Their Connectors
BS-EN 13599 - 2002	Copper and copper alloys. Copper plate, sheet and strip for electrical purposes
BS-EN 13601 - 2002	Copper and copper alloys. Copper rod, bar and wire for general electrical purposes
BS-EN1982	Copper and Copper alloys. Ingots and Castings

4.2 Internal documentation

Reference	Title
IMP/010/011	Code of Practise for Earthing LV Networks and HV Distribution Substations.

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4.3 Amendments from Previous Version

Clause	Subject	Amendments
2	Scope	Ref to CoP IMP/010/011 and Energy Networks Technical Association Specification 43-94
2.1	Contents	Table of contents added.
3.4 / Appdx 1	Earth Plate Assembly	Item deleted
4.1	External Documents	BS 4620 - Specification for rivets for general engineering purposes deleted.
4.2	Internal Documents	Reference to Code of Practice for Earthing LV Networks and HV Distribution Substations added.
4.3	Amendments	Amendments from previous version table added.
Appdx 4	Self Declaration	Self declaration tables added

5. Definitions

Term	Definition
NIL	

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6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
V Gorny	Standards Engineer	V Gorny	16/03/2010

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
G Bolden	Procurement Manager	G Bolden	17/03/2010

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
R McMahon	Standards Engineer	R McMahon	16/03/2010

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S Johnson	DBD Administrator	S Johnson	19/03/2010

6.5 Approval

Approval is given for the content of this document

		Sign	Date
C. Holdsworth	Standards and Assessment Manager	C. Holdsworth	17/03/2010

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
C. Morgan	Head of Procurement	C. Morgan	17/03/2010

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Appendix 1 – Requirements and Catalogue Numbers

DESCRIPTION	CE Electric UK Catalogue Number
Earth Rod: (Extensible) 1.2 M Nominal Length & 16mm Nominal Diameter (Typical arrangement - Drawing 1.09.119.3143 Sheet 1)	354249
Earth Rod Coupling: (For Extensible earth rod) (Typical arrangement - Drawing 1.09.119.3143 Sheet 1)	354361
Steel Driving Stud: (For extensible earth rod) (Typical arrangement - Drawing 1.09.119.3143 Sheet 1)	354408
Earth Rod Connector: (two types required): Cable clamp type: Typical arrangement - Drawing 1.09.119.3143 Sheet 2) & Bulldog Grip Type: Typical arrangement - Drawing 1.09.101.0383 Sheet 1)	354338
	222398
High Conductivity Plain Copper Strip: 4mm x 25mm	331213
4mm x 40mm	281832
Flat Aluminium Strip: (60 x 4 0mm)	
Plain Copper Braid	
2mm x 24mm x 37 x 0.15mm (50 Amp)	330672
3mm x 32mm x 22 x 0.2mm (100 Amp)	330704

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Appendix 2 – Addendum to Supplier Requirements

The supplier shall provide with the tender full technical details of the equipment offered and shall indicate any variation from these standards or specifications.

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Appendix 3 – Logistical Requirements

To enable CE Electric UK to store the product(s) in accordance with the manufacturer's recommendations the Tenderer should provide details of the recommended storage environment with respect to each tendered product.

Details should be provided where relevant in respect to the minimum and maximum exposure levels, frequency of exposure and duration of exposure of the packaged item with respect to;

- | | | | |
|---|---------------------|---|-----------------------|
| * | Ambient temperature | * | Atmospheric corrosion |
| * | Humidity | * | Impact |
| * | Water | * | Vibration |
| * | Dust | * | Solar radiation |

The Tenderer shall ensure that each item is suitably packaged and protection to maintain the product and packaging as "fit for service" prior to installation taking account of the potential for an outdoor storage environment. All packaging shall be sufficiently durable giving regard to the function, reasonable use and contents of the packaging. Where product packages tendered are made up of sub packages all the sub packages shall unless varied by this specification, be supplied securely packaged together. Where items are provided in bagged/boxed form the material from which the bags are manufactured shall be capable of sustaining the package weight and resisting puncture by the materials within. Tenderer shall submit at the time of tendering the details of the proposed packaging (i.e. materials composition and structure) to be used for each product. Where the Tenderer is unable to provide packaging suitable for outdoor storage then this should be stated at the time of tender.

Palletised goods shall be supplied on standard 1200mm x 1000mm pallets.

Clearly legible, easily identifiable, durable and unambiguous labelling shall be applied to each individual and where relevant multiple packages of like products. Where product packages tendered are made up of sub packages each sub packages shall be marked. As a minimum requirement the following shall be included;

- * Manufacturer's trademark or name
- * Supplier's trademark or name
- * Description of item
- * Date of packaging and/or batch number
- * CE Electric UK product code
- * Weight

Tenderer shall submit at the time of tendering a sample of the proposed labelling for each product package type.

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Appendix 4 – SELF CERTIFICATION CONFORMANCE DECLARATION

The Technical Specification for Earthing Materials shall comply with the latest issues of the relevant national and international standards, including ENATs 43-94.

Additionally this technical specification is intended to amplify and/or clarify requirements relating to these Standards.

This self declaration sheet identifies the clauses of the aforementioned standards relevant to Earthing Materials for use on the CE Electric UK distribution network. The manufacturer shall declare conformance or otherwise, clause by clause, using the following levels of conformance declaration codes.

Conformance declaration codes

N/A = Clause is not applicable/ appropriate to the product

Cs1 = The product conforms fully with the requirements of this clause

Cs2 = The product conforms partially with the requirements of this clause

Cs3 = The product does not conform to the requirements of this clause

Cs4 = The product does not currently conform to the requirements of this clause, but the manufacturer proposes to modify and test the product in order to conform.

Instructions for completion

- When Cs1 code is entered no remark is necessary
- When any other code is entered the reason for non- conformance shall be entered
- Prefix each remark with the relevant 'BS EN' 'IEC' or 'ENATS' as appropriate.

Manufacturer:

Product Reference:

Name:

Signature:

Date:

NOTE: One sheet shall be completed for each item or variant submitted.

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Appendix 4 Cont'

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• <u>ENATS 43-94</u>			
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4.1	Fabrication and Marking		
5	Components		
5.1	Earth Rods		
5.2	Driving Heads		
5.3	Driving Tip		
5.4	Couplers		
5.4.1	General		
5.4.3	Requirements for Dowels for Copper Bonded Rod		
5.4.4	Couplers of Alternate Design		
5.5	Connectors		
6	Testing		
6.1	General		

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• <u>ENATS 43-94</u>			
Clause/Sub-clause	Requirement	Conformance Code	Remarks
6.2	Type Tests		
6.2.1	Sample Size		
6.2.2	Visual Examination		
6.2.2.1	Requirements for Copper Bonded Rods		
6.2.2.3	Requirements for Galvanised Couplers and Driving Tips		
6.2.2.4	Requirements for Sherardized Couplers		
6.2.3	Rod Bend Test		
6.2.4	Adhesion Test (Copper Bonded Earth Rod)		
6.3	Sample Tests		
6.3.1	Sample Size		
6.3.1	Sample Tests		

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Clause/Sub-clause	Requirement	Conformance Code	Remarks
3.5 High Conductivity Plain Copper Strip	BS EN 13601 (embossed every 300mm with the wording "Property of CE Electric UK").		
3.6 Flat Aluminium Strip.	BS 2898 – 1350 embossed every 300mm with the wording "Property of CE Electric UK".		
3.7 Plain Copper Braid.	BS EN 13602.		